Internship Report

Of

NETSAGE AI: Artificial Intelligence and Network Troubleshooting

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number : CS-23411216

Name of Student: ANKESH KUMAR

Batch: 2023-27

2026-27

CANDIDATE'S DECLARATION

I, ANKESH KUMAR do hereby declare that the internship report titled " NETSAGE AI: Artificial Intelligence and Network Troubleshooting " has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents and everyone who encouraged me throughout the completion of this project and report.

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Student Name: Ankesh Kumar

Roll No.: CS-23411216

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3. INTERNSHIP COMPLETION CERTIFICATE

The internship completion and course-certification evidence supplied for this report is reproduced in Appendix C. The three certificates confirm successful completion of Introduction to Modern AI, Data Analytics Essentials, and Apply AI: Analyze Customer Reviews through IILM University and the Cisco Networking Academy program.

All three certificates were issued in the name of SANDEEP KUMAR with a completion date of 15 June 2026. The certificates are intentionally placed at the end of the report so that the final report follows the submission arrangement specified in the updated internship-report format.

Course	Provider / Platform	Completion Date	Certificate ID
Introduction to Modern AI	IILM University / Cisco Networking Academy	15 Jun 2026	76e68725-a745-4d22-8163-3a59b232d645
Data Analytics Essentials	IILM University / Cisco Networking Academy	15 Jun 2026	96cc0395-fbc4-4031-aea2-ac7ee034c86c
Apply AI: Analyze Customer Reviews	IILM University / Cisco Networking Academy	15 Jun 2026	24216727-5248-4fb2-88ab-7abcfbdbb804

4. PROJECT DESCRIPTION

4.1 Introduction

NetSage AI — Network Insight Edition is a browser-based decision-support project for Cisco-style network troubleshooting. The project was designed to organize common network incidents into a structured dataset and present evidence-based troubleshooting suggestions through a clean dashboard.

The project contains 30 predefined troubleshooting cases covering VLANs, routing, DHCP, DNS, ACLs, NAT, wireless networking, interfaces, trunks, OSPF, static routes, subnetting, and related concepts. The dashboard provides an overview of case patterns, severity, OSI-layer distribution, and a case explorer. A separate AI Troubleshooting workspace uses a deterministic rule-based engine to infer a likely root cause from the supplied symptom and evidence.

A key design principle is responsible human oversight. The system presents recommendations as advisory, provides a verification command, and requires human review before any configuration change. This makes the project appropriate as an academic laboratory and decision-support demonstration rather than an autonomous production-network controller.

4.2 Organization Profile

IILM University, Greater Noida, U.P., provides the academic environment for the B.Tech CSE internship requirement. The learning evidence supplied with the internship report shows coursework delivered through the Cisco Networking Academy program. The documented learning path covered modern AI, data analytics, and applied AI, creating a foundation for the NetSage AI project.

The project connects these learning areas with a practical technical application. Data organization and evaluation are represented through structured CSV files; AI concepts are reflected in the evidence-to-diagnosis workflow; visualization skills are demonstrated through the dashboard; and responsible-AI concepts are implemented through confidence reporting and a human-review gate.

4.3 Problem Statement

Network troubleshooting often requires moving through several reasoning steps: recognizing the symptom, examining command output or host evidence, identifying a probable fault domain, selecting a verification command, and deciding whether a configuration change is justified. For learners and junior operators, this sequence can be difficult to apply consistently.

A troubleshooting assistant that immediately provides a configuration command can also encourage premature changes. NetSage AI addresses this problem by showing the evidence used for a diagnosis, presenting a probable root cause with confidence, recommending a verification command, and requiring human approval before a configuration change.

4.4 Project Objectives

- Create a clear browser-based dashboard for Cisco-style troubleshooting cases.
- Maintain a structured dataset of 30 network incidents with symptoms, evidence, expected faults, OSI layers, concepts, and severity.
- Provide rule-based AI-assisted diagnosis using the supplied symptom and evidence.
- Recommend a diagnostic command before a configuration change.
- Provide generic and safe troubleshooting steps that require human approval.
- Use a deterministic evidence checker to identify common configuration clues independently of the diagnosis engine.
- Record selected human acceptance or editing decisions through a responsible-AI review log.
- Present the project through a clean, professional purple dashboard suitable for demonstration and academic evaluation.

4.5 Scope of the Project

The project scope is intentionally focused on a controlled Cisco-style laboratory dataset and a local browser dashboard.

In scope:

- Cisco-style network troubleshooting and evidence interpretation.
- Thirty predefined cases from the supplied dataset.

- Interactive case exploration and structured diagnosis.
- Severity and OSI-layer summaries.
- Human review logging and responsible-AI messaging.
- Dataset-level testing of diagnosis output.

Out of scope:

- Autonomous configuration changes on routers, switches, wireless controllers, or production systems.
- Live device telemetry or real-time network monitoring.
- A general-purpose machine-learning model trained on live network traffic.
- Guaranteeing diagnosis correctness outside the supplied dataset.

4.6 Technologies and Tools Used

Component	Technology / Approach	Purpose
Dashboard	HTML5 + CSS + JavaScript	Interactive browser interface
Dashboard Generator	Python (dashboard.py)	Loads CSV data and generates dashboard.html
Diagnostic Engine	Python (ai.py)	Rule-based evidence-to-diagnosis mapping
Evidence Checker	Python (checker.py)	Deterministic checks for common evidence patterns
Dataset	cases.csv	30 structured troubleshooting cases
Review Log	responsible_ai_log.csv	Human review records
Diagnosis Guidance	diagnose_prompt.md	Evidence-first diagnosis and safety guidance
Execution	Python 3 + Web Browser	Runs the project locally

The implementation intentionally uses transparent, inspectable components. The report refers to the implemented engine as AI-assisted decision support; the underlying diagnostic logic is deterministic and rule based rather than a trained neural network.

4.7 System Architecture

The system follows a simple local architecture. Structured case data and review records are loaded from CSV files. The diagnostic engine evaluates the selected case using ordered evidence rules. The dashboard generator combines the data and diagnosis results into a standalone HTML dashboard. Human reviewers can then inspect and refine selected recommendations.

Stage	Input	Processing	Output
1. Load data	cases.csv / review log	Python CSV parsing	Structured case and review records
2. Diagnose	Symptom + evidence	Ordered keyword rules	Root cause + confidence + next command
3. Validate	Case evidence	Deterministic checker	Evidence findings
4. Present	Case + diagnosis	HTML/CSS/JS rendering	Dashboard and case explorer
5. Review	AI recommendation	Human reviewer	Accepted / Edited / Rejected record
6. Evaluate	All 30 cases	Token-overlap criterion	Dataset-level match percentage

The evidence-first workflow is: observe the symptom, inspect the supplied evidence, identify the most likely fault, assign confidence, recommend a verification command, provide generic fix steps, require human review, and retest the original symptom after an approved change.

4.8 Methodology

1. Data preparation: organize network incidents into structured case records.
2. Evidence interpretation: combine the case symptom and supplied evidence for analysis.
3. Rule-based diagnosis: evaluate ordered clues, with more specific rules placed before generic fallbacks.
4. Independent validation: run deterministic evidence checks for common configuration patterns.
5. Dashboard presentation: display summaries, cases, diagnosis information, and review records in a clean interface.
6. Human review: require a reviewer to confirm, edit, or reject advisory recommendations.
7. Evaluation: compare generated root causes against expected faults using the project's explicit dataset-level matching routine.

4.9 Expected Outcomes

- A readable and professional network troubleshooting dashboard.
- Structured visibility into issue type, severity, and OSI-layer distributions.
- Evidence-based root-cause hypotheses with confidence and verification commands.
- An auditable human-review mechanism for AI-assisted recommendations.
- Measurable dataset-level evaluation of the implemented diagnostic rules.
- Practical learning in Python, HTML/CSS/JavaScript, structured data, networking, dashboard design, and responsible AI.

5. PROJECT IMPLEMENTATION AND DASHBOARD

5.1 Dataset and Case Design

The supplied dataset contains exactly 30 troubleshooting cases. Each row includes a case ID, issue type, symptom, evidence, expected fault, OSI layer, concept, and severity. The cases intentionally represent multiple layers and fault categories so that the dashboard can provide a broad view of network troubleshooting.

Issue Type	Cases
ACL	4
Routing	3
VLAN	3
Wireless	3
DHCP	2
DNS	2
Interface	2
NAT	2
DHCP Relay	1
Duplicate IP	1
Gateway	1
Mask	1
Native VLAN	1
OSPF	1
SVI	1
Static Route	1
Trunk	1

5.2 Severity and OSI Distribution

Severity	Count	Share
Critical	1	3.3%
High	17	56.7%

Medium	12	40.0%

OSI Layer	Count
Layer 3	15
Layer 2	5
Layer 4	3
Layer 1/2	2
Layer 2/3	2
Layer 7	2
Layer 3/4	1

5.3 Dashboard Design

The redesigned dashboard uses a dark navy navigation white content cards, a light-gray page background, rounded panels, compact severity indicators, and consistent spacing. The design is intended to be clearer and cleaner than a dense troubleshooting table while retaining the project's functionality.

The main workspaces are Dashboard, AI Troubleshooting, and Review Centre. The Dashboard provides high-level KPIs and distribution summaries. The AI Troubleshooting screen presents a selected case, evidence, root cause, confidence, and verification command. The Review Centre shows review statistics and human-review records.

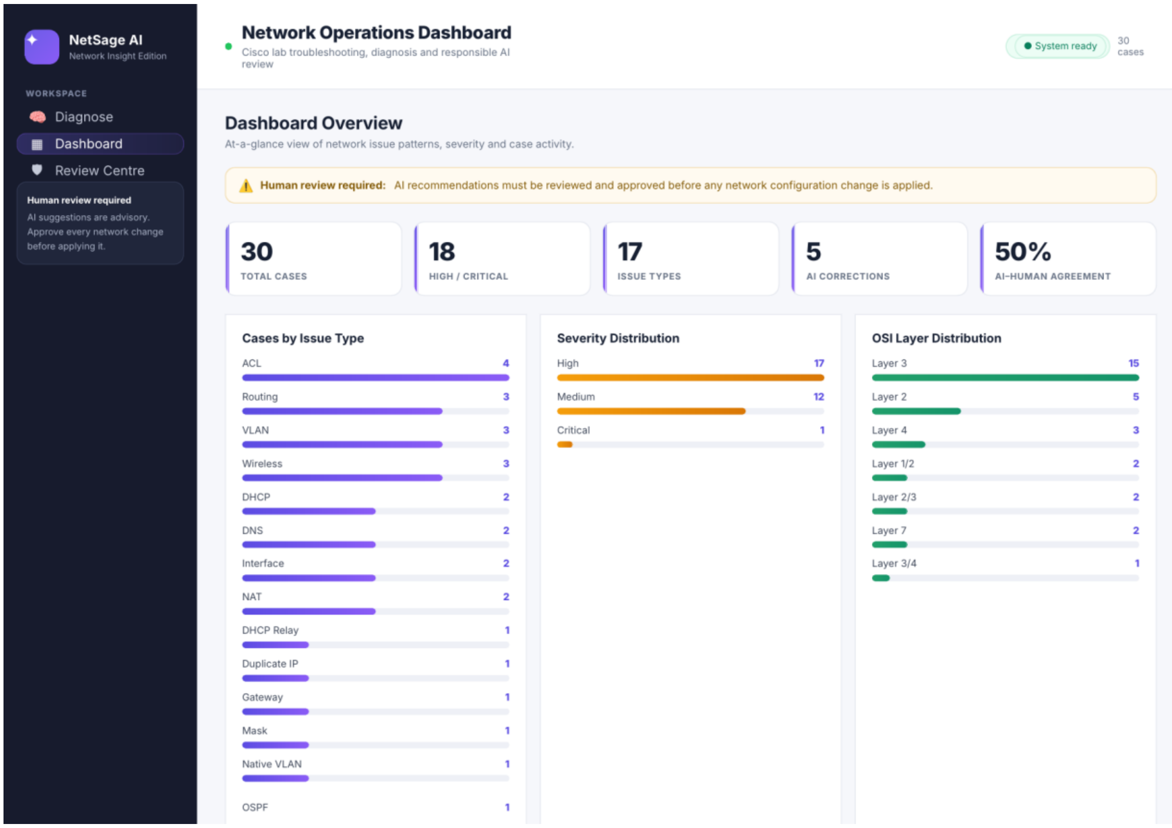

Figure 5.1. NetSage AI dashboard overview showing total cases, severity, issue-type distribution, OSI layers, and the human-review warning.

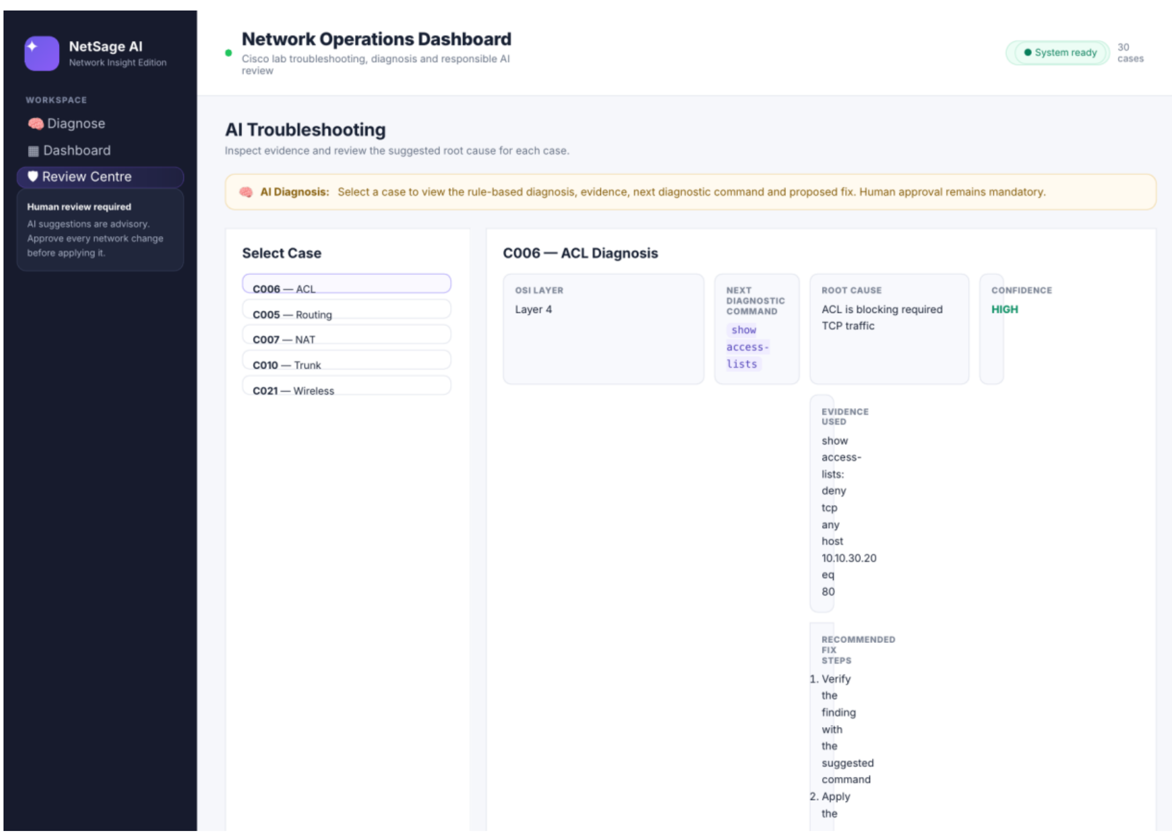

Figure 5.2. AI Troubleshooting screen showing a structured diagnosis for case C006 and the required human-review workflow.

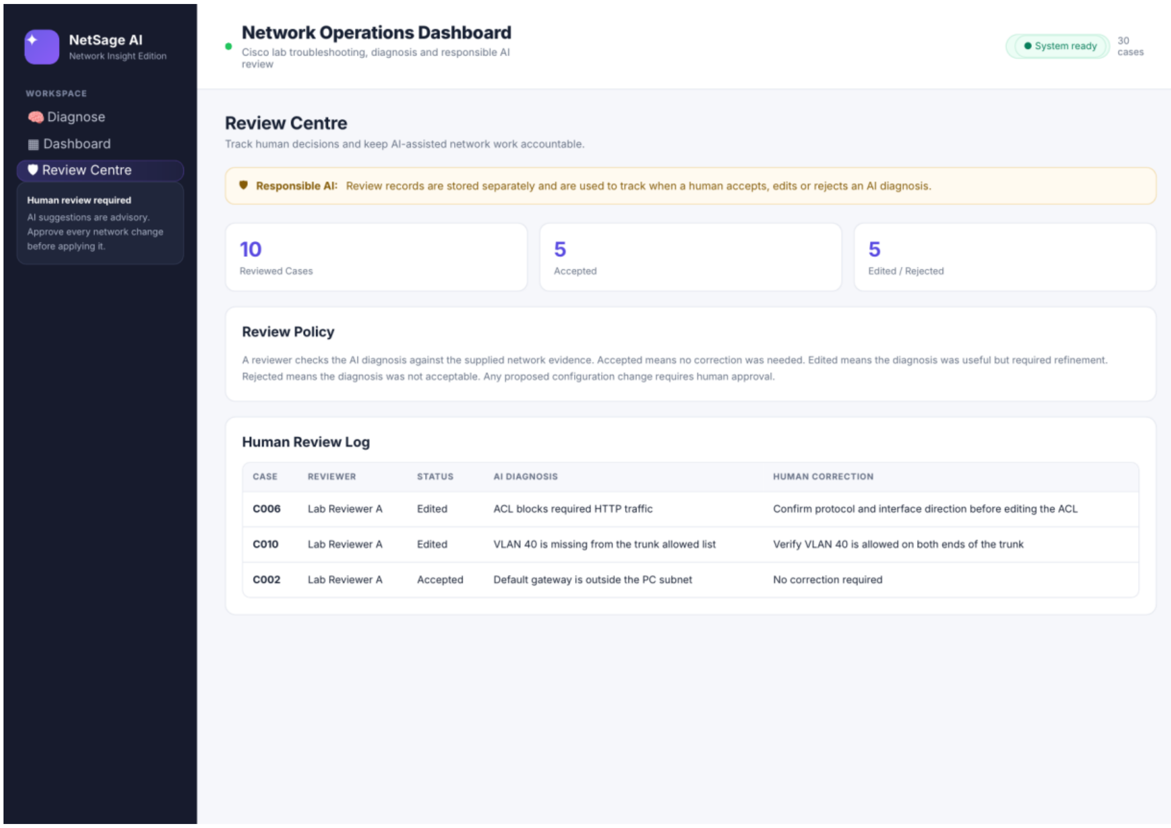

Figure 5.3. Review Centre showing review KPIs, review policy, and human-review records.

5.4 AI Diagnostic Engine

The diagnostic engine in ai.py is deterministic and rule based. It combines the supplied evidence and symptom, converts the combined text to a consistent form, and checks an ordered list of clues. When a rule matches, the engine returns the most likely root cause, high confidence, the case OSI layer, the evidence used, a recommended next command, generic fix steps, and a required human-review flag.

In this project, the term AI refers to the AI-assisted decision-support workflow and structured diagnostic behavior. The implemented engine is not a trained neural model. This distinction is important for academic accuracy and for understanding the project's limitations.

5.5 Deterministic Evidence Checker

checker.py provides a second validation layer. It looks for common evidence patterns such as administratively down interfaces, missing routes, gateway mismatches, absent VLANs, native VLAN mismatches, missing DHCP relay addresses, exhausted DHCP pools, ACL deny rules, NAT mismatches, and speed/duplex problems.

Evidence Pattern	Example Finding
administratively down	Interface appears administratively down.
missing route / no route	A routing entry appears to be missing.
gateway mismatch	Default gateway does not match the host subnet/design.
VLAN absent	Required VLAN appears to be missing.
no ip helper-address	DHCP relay/helper address is missing.
pool exhausted	DHCP address pool is exhausted.
deny tcp / deny ip	ACL contains a relevant deny rule.
wrong outside / wrong global	NAT configuration contains a likely mismatch.
speed/duplex	Speed/duplex mismatch is indicated.

5.6 Responsible AI and Human Review

Responsible AI is integrated into the project rather than treated as an optional interface element. Every diagnosis carries a human-review requirement, and the dashboard warns that recommendations are advisory. The engine does not execute configuration changes.

Review Status	Count	Interpretation
Accepted	5	No correction required by reviewer.
Edited	5	Diagnosis was useful but required refinement.
Rejected	0	No rejected records in the supplied log.

The ten reviewed records demonstrate why human review remains important. Edited examples include ACL direction, trunk-path verification, next-hop reachability, route administrative distance, and management-interface confirmation. Accepted examples include clear gateway, DNS, NAT, duplicate-IP, and wireless VLAN findings.

6. TESTING, RESULTS AND DISCUSSION

6.1 Testing Approach

The project includes a dataset-level evaluation routine in ai.py. For each case, the expected fault and generated root cause are tokenized and compared using the project's explicit overlap criterion. A case is counted as a match when the overlap reaches the implemented threshold.

This evaluation measures performance against the supplied controlled dataset. It should not be interpreted as a benchmark of real-world network diagnostic accuracy.

6.2 Evaluation Result

Metric	Result
Total cases evaluated	30
Matching cases	28
Non-matching cases	2
Dataset-level match	93.3%
High/Critical cases	18
Human reviews	10
Accepted reviews	5
Edited reviews	5
Rejected reviews	0

The implemented rule set matched 28 of 30 cases, giving a dataset-level match of 93.3%. Cases C016 and C029 produced low-confidence results because the current rule wording did not match the wording used in the supplied evidence. This demonstrates a practical limitation of keyword-based systems: they can perform strongly on controlled phrasing while remaining sensitive to vocabulary variation.

6.3 Cases Requiring Improvement

Case	Issue	Expected Fault	Current Result	Reason

C016	Static Route	Incorrect next-hop address	Low confidence / no match	Current keyword rule did not match the dataset wording.
C029	Wireless	SSID mapped to wrong VLAN	Low confidence / no match	Current keyword rule did not match the dataset wording.

6.4 Representative Case-wise Evaluation

ID	Issue	Severity	Expected Fault	Result
C001	VLAN	High	VLAN 20 missing	Match
C002	Gateway	High	Gateway outside PC subnet	Match
C003	DHCP	High	Wrong DHCP pool network	Match
C004	DNS	Medium	Incorrect DNS server	Match
C005	Routing	High	Missing route to remote LAN	Match
C006	ACL	High	ACL blocks HTTP	Match
C007	NAT	High	Wrong outside interface	Match
C008	Wireless	Medium	Missing default route	Match
C009	Interface	High	Interface administratively down	Match
C010	Trunk	High	VLAN 40 absent from allowed list	Match
C011	Mask	Medium	Incorrect subnet mask	Match
C012	Duplicate IP	High	Duplicate IP / ARP instability	Match
C013	SVI	High	Subinterface line protocol down	Match
C014	Native VLAN	Medium	Native VLAN mismatch	Match
C015	OSPF	High	Remote network not advertised	Match
C016	Static Route	High	Incorrect next hop	No match
C017	ACL	Medium	SSH restricted to one source	Match
C018	DHCP Relay	High	Missing helper address	Match
C019	DNS	Medium	Stale DNS record	Match
C020	NAT	High	Incorrect translation rule	Match

C021	Wireless	Critical	Missing guest-isolation ACL	Match
C022	VLAN	Medium	Missing voice VLAN	Match
C023	Interface	Medium	Speed/duplex mismatch	Match
C024	Routing	High	Incorrect backup-route distance	Match
C025	ACL	Medium	HTTPS blocked while ICMP allowed	Match
C026	DHCP	Medium	DHCP pool exhausted	Match
C027	VLAN	Medium	Wrong switch-port VLAN	Match
C028	Routing	Medium	More-specific route precedence	Match
C029	Wireless	High	SSID mapped to wrong VLAN	No match
C030	ACL	High	Management subnet denied	Match

6.5 Sample End-to-End Diagnosis

Field	Value
Case	C006 — ACL / HTTP Access
Symptom	Users can reach the gateway but HTTP traffic to the server times out.
Evidence	show access-lists: deny tcp any host 10.10.30.20 eq 80
Expected Fault	ACL blocks required HTTP traffic
AI Root Cause	ACL is blocking required TCP traffic
Confidence	High
OSI Layer	Layer 4
Next Diagnostic Command	show access-lists
Human Review	Required

The evidence explicitly contains a TCP deny rule for HTTP traffic to the server. The engine therefore identifies the ACL as the likely cause and recommends checking the access list. The project does not automatically remove or modify the ACL. A human reviewer should confirm the protocol and interface direction, approve any change if appropriate, and retest the original application symptom.

7. LEARNING OUTCOMES, LIMITATIONS AND FUTURE SCOPE

7.1 Learning Outcomes

- Applied AI concepts to a concrete network troubleshooting decision-support workflow.
- Practiced structured data organization using CSV-based case and review records.
- Applied evidence-driven reasoning instead of unsupported assumptions.
- Connected network troubleshooting concepts to OSI layers and common Cisco diagnostic commands.
- Used HTML, CSS, and JavaScript to present technical information in a clear dashboard.
- Applied responsible-AI principles through confidence reporting and mandatory human review.
- Used evaluation results to identify specific engineering limitations and future improvement opportunities.
- Improved understanding of how data analytics, visualization, and AI-assisted workflows can work together in a practical project.

7.2 Limitations

- Rule-based keyword matching is sensitive to wording; C016 and C029 demonstrate this limitation.
- The current dataset contains controlled evidence rather than live network telemetry.
- The diagnostic engine does not model full topology relationships or configuration dependencies.
- The evaluation metric is token overlap against expected faults and is not a real-world accuracy benchmark.
- The review log is a static CSV rather than a multi-user audit database.
- The project is a local decision-support demonstration and does not provide autonomous production-network control.

7.3 Future Enhancements

- Replace brittle keyword matching with normalized evidence extraction and synonym handling.

- Add topology-aware reasoning for interfaces, VLANs, trunks, routing paths, and next-hop reachability.
- Support uploaded Cisco show-command files and structured evidence parsing.
- Add filters and search by severity, OSI layer, concept, and issue type.
- Add reviewer authentication, timestamps, and immutable audit records.
- Add precision, recall, confusion analysis, and per-category evaluation metrics.
- Introduce retrieval or machine-learning components while retaining the deterministic checker and human approval gate.

8. CONCLUSION

NetSage AI — Network Insight Edition demonstrates how a small, transparent AI-assisted system can organize Cisco-style troubleshooting into a repeatable evidence-first workflow. The dashboard makes network case patterns easier to inspect, the diagnostic engine provides a structured hypothesis and verification command, the deterministic checker provides an independent evidence layer, and the Review Centre keeps a human decision-maker in the loop.

The current implementation matched 28 of 30 supplied cases, or 93.3%, under its explicit dataset-level evaluation criterion. The two non-matching cases are useful engineering evidence of the limits of keyword-based diagnosis and provide clear targets for future enhancement.

Overall, the project provides a practical academic demonstration of applied AI, network troubleshooting, dashboard design, structured data handling, testing, and responsible human oversight. It also demonstrates how the learning themes documented during the internship can be connected to a concrete software project.

9. BIBLIOGRAPHY / REFERENCES

- NetSage AI project package: README.md, ai.py, checker.py, dashboard.py, dashboard.html.
- NetSage AI dataset: cases.csv.
- NetSage AI responsible-AI records: responsible_ai_log.csv.
- NetSage AI diagnosis guidance: diagnose_prompt.md.
- IILM University / Cisco Networking Academy — Introduction to Modern AI certificate evidence.
- IILM University / Cisco Networking Academy — Data Analytics Essentials certificate evidence.
- IILM University / Cisco Networking Academy — Apply AI: Analyze Customer Reviews certificate evidence.
- Internship Report Format updated 26–27, School of Computer Science and Engineering, IILM University.

APPENDIX A — PROJECT FILE STRUCTURE

File	Role
ai.py	Rule-based diagnostic engine and dataset-level evaluation
checker.py	Deterministic evidence checker
dashboard.py	Dashboard generation and aggregation
dashboard.html	Interactive clean purple dashboard
cases.csv	30 troubleshooting cases
responsible_ai_log.csv	Human review records
diagnose_prompt.md	Structured evidence-first diagnosis and safety guidance
checker_sample_output.txt	Sample checker output

APPENDIX B — DASHBOARD SCREENS

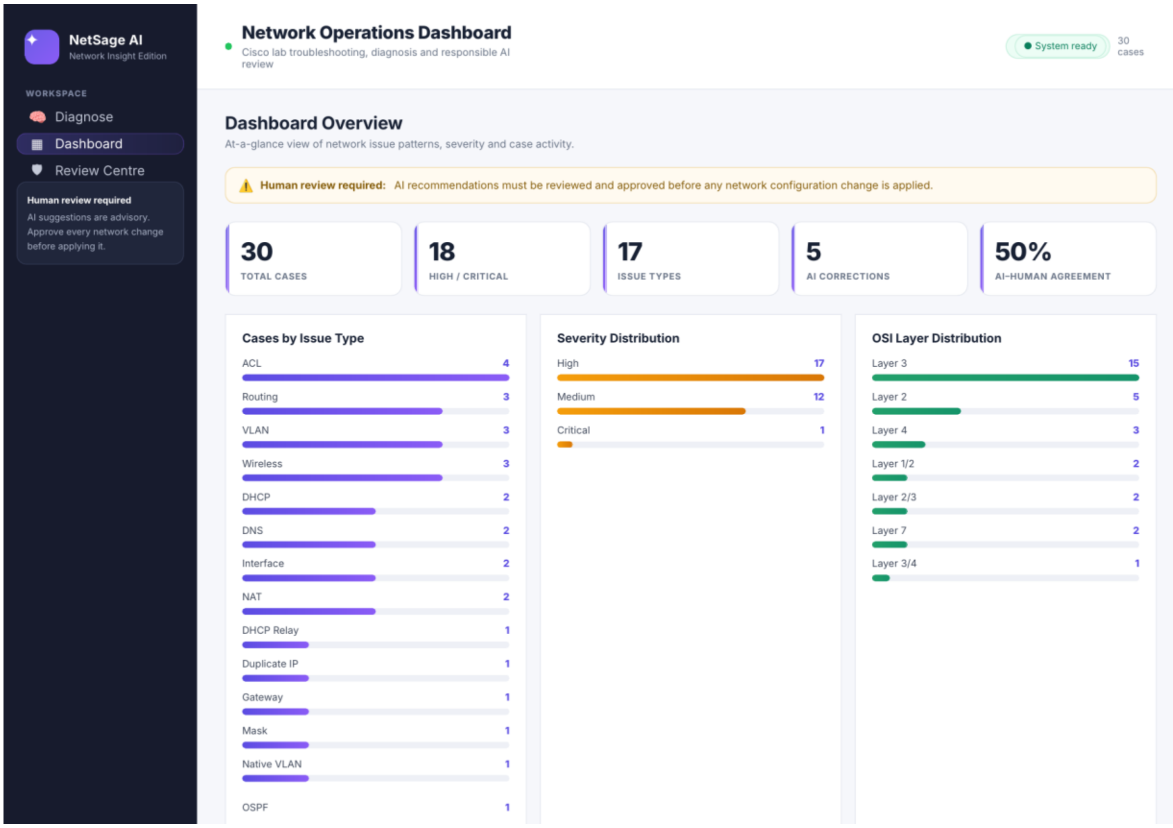

Figure B.1. Dashboard Overview.

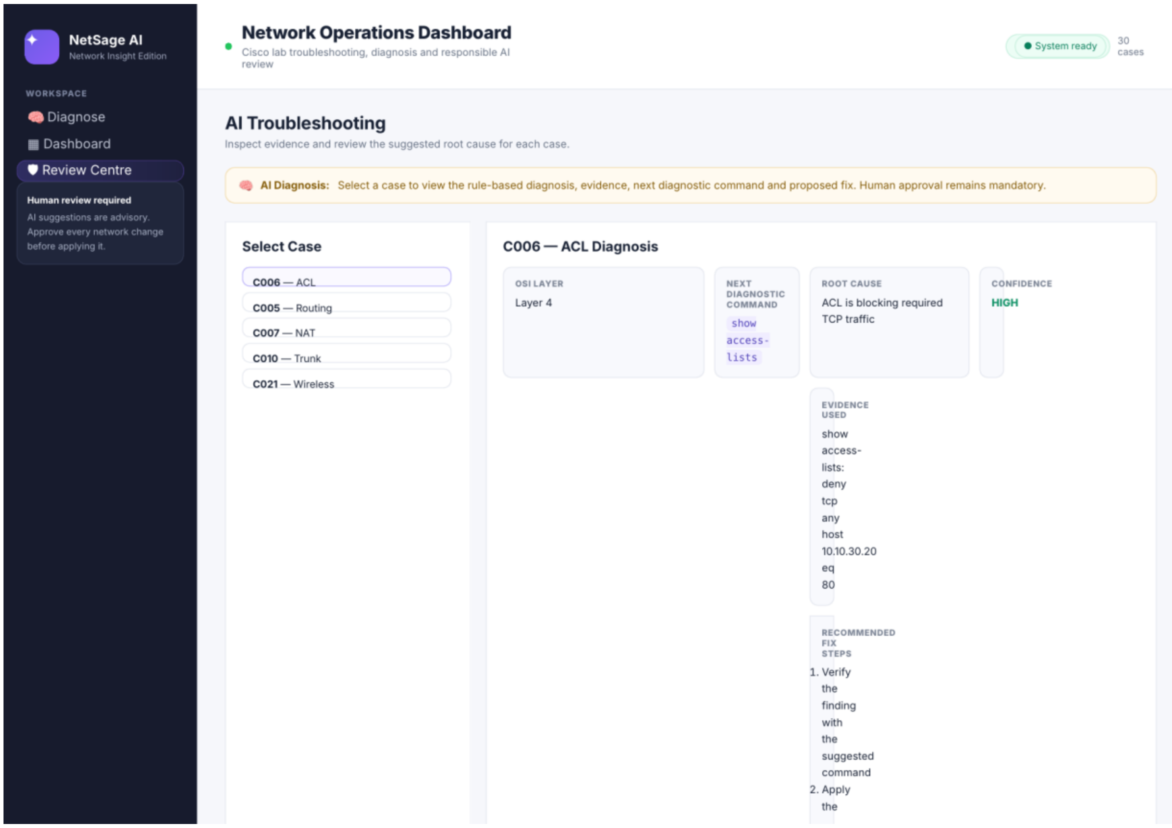

Figure B.2. AI Troubleshooting / Diagnosis.

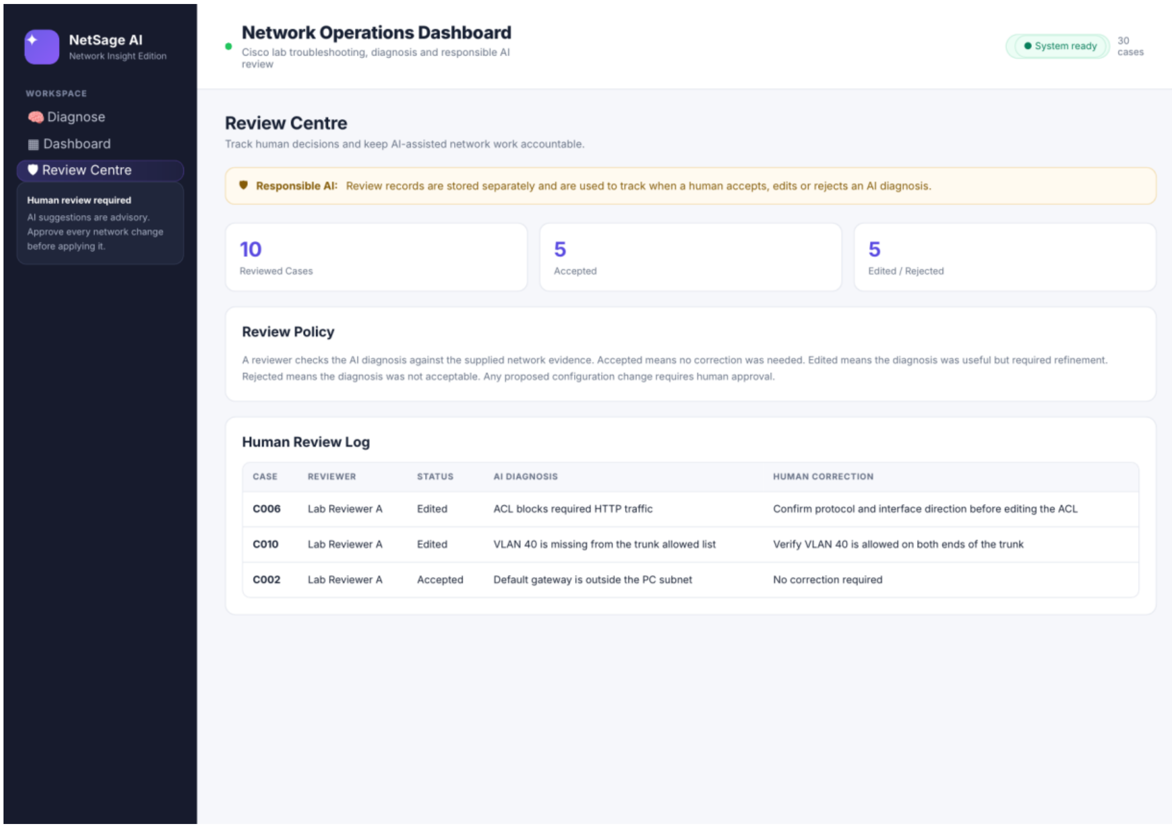

Figure B.3. Responsible AI Review Centre.

APPENDIX C — INTERNSHIP CERTIFICATES

The following three certificate pages are reproduced from the certificate evidence supplied for this internship report. They are placed at the end of the report in accordance with the internship-report submission arrangement.

Certificate 1 — Apply AI: Analyze Customer Reviews

Cisco
Networking
Academy

IILM
UNIVERSITY

This certificate is awarded to

Ankesh kumar

for successfully completing

Apply AI: Analyze Customer Reviews

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht
Instructor
IILM University

15 Jun 2026
Completion Date

Cert ID: 06a8ce98-faf3-4448-99a8-266c7f972663

Certificate 2 — Introduction to Modern AI

Cisco
Networking
Academy

IILM
UNIVERSITY

This certificate is awarded to

Ankesh kumar

for successfully completing

Introduction to Modern AI

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht
Instructor
IILM University

14 Jun 2026
Completion Date

Cert ID: fb9b5a54-5e43-40c2-a393-4d7c8faba728

Certificate 3 — Data Analytics Essentials

Cisco
Networking
Academy

IILM
UNIVERSITY

This certificate is awarded to

Ankesh kumar

for successfully completing

Data Analytics Essentials

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht
Instructor
IILM University

11 Jun 2026
Completion Date

Cert ID: 73223362-cd6c-4fdd-b426-aa7b2a0f0ca3